Soumya Mazumdar

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RESEARCH STATEMENT

My research asks: *How can sparse human-authored structure (strokes, line-art, landmarks, and pose) be lifted into temporally coherent, controllable video generation?* I engineer geometry-aware generative models that factorize **motion** from **appearance**, then enforce temporal consistency through explicit correspondences such as landmark or pose trajectories, learned point tracks, and optical flow, together with optimization-driven objectives. Motivated by artist-centric animation pipelines, I am particularly interested in **temporal colorization for sketch/line-art**: conditioning diffusion or transformer models on line structure and sparse color hints while preserving edges, identity, and long-range color stability. This aligns with the DynColor direction–assistive colorization that behaves predictably across frames and respects drawn geometry.

RESEARCH FOCUS

Temporal Generative Modeling: diffusion for video; long-range temporal consistency **Sketch/Line-Art Priors:** artist-in-the-loop control; sparse hints; controllable synthesis
Geometry in Vision: landmarks/keypoints; head pose; projection-aware conditioning **Optimization Mindset:** rigorous ablations; reproducible benchmarks; deployment-aware speedups

RESEARCH EXPERIENCE

Variable Energy Cyclotron Centre (DAE, Government of India) Mar. 2025 – Apr. 2026
Research Trainee

- Researching **multimodal spatio-temporal generative models** for human communicative behavior modeling. Worked on synchronization, temporal consistency, and geometry conditioning in audio-visual generation pipelines. Built reproducible training pipelines using **PyTorch** and **CUDA** for multimodal sequential generation and evaluation.
- Co-authored research works published, accepted, or under review in Springer, Scientific Reports, IEEE, and Wiley

LEADERSHIP & ROLES

IgniteSpark R&D Cell, GMIT – *Vice President* 2023 – 2025
Directed a 15-member R&D team; standardized experiment tracking and peer review across projects.

Dhyan Chand Sports Council, IIT Madras – *Eastern Zonal Secretary* 2023
Coordinated multi-state student participation and technical recruitment drives involving **100+ candidates**.

TECHNICAL SKILLS

Programming: Python, C++, MATLAB
Frameworks: PyTorch, TensorFlow, Keras
CV/Video: OpenCV, MediaPipe, Decord
3D/Graphics: Blender, Open3D, Unity (basic)
Systems: Linux, Git, CUDA, LaTeX
Deep Learning: Diffusion Models, GANs, VAEs, Vision Transformers, Conformer-CTC
Math: Linear Algebra, Probability & Statistics; geometry priors (keypoints/pose)

EDUCATION

Maulana Abul Kalam Azad University of Technology *(Gargi Memorial Institute of Technology campus)* Nov. 2023 – Jun. 2027
B.Tech, Computer Science & Business Systems (Hons)

Indian Institute of Technology Madras May 2023 – Apr. 2027
BS, Data Science and Applications

HONORS & AWARDS

Anusandhan National Research Foundation (ANRF) ITS Funding for the ICIR 2026 Conference [File No. ITS/2026/002133]
Swami Vivekananda Merit-cum-Means Scholarship
Kendriya Sainik Board Scholarship

LANGUAGES

English (Full Professional) | Hindi (Full Professional) | Bengali (Native)

REFERENCES

Dr. Vineet Kumar Rakesh – Senior Technical Officer (Scientific), Variable Energy Cyclotron Centre, Department of Atomic Energy, Government of India; vineet@vecc.gov.in; [Reference link].
Dr. Somnath Maiti – Principal, Gargi Memorial Institute of Technology; principal_gmit@jisgroup.org; [Reference link].

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RESEARCH PORTFOLIO & SCHOLARLY WORKS

Journal Articles • Patent • Conference Papers • Book & Chapters • Preprints

JOURNAL ARTICLES

1. **Rakesh, V. K., Mazumdar, S., Maity, R. P., Pal, S., Das, A., & Samanta, T.** (2026). Advancing Talking Head Generation: A Comprehensive Survey of Multi-Modal Methodologies, Datasets, Evaluation Metrics, and Loss Functions. *The Visual Computer*, **42**(9). doi:10.1007/s00371-025-04232-w.
2. **Rakesh, V. K., Mazumdar, S., Samanta, T., Pandey, H. K., Das, A., & Pal, S.** (2026). Analysis of Hyperparameter Optimization Effects on Lightweight Deep Models for Real-Time Image Classification. *Scientific Reports*. doi:10.1038/s41598-026-42748-w.

PATENT

3. **Mazumdar, S., Saha, S., Roy, M., Kundu, S., Halder, P. K., Das, S., Adhikari, U., & Nayak, L.** (2026). *Handheld Device for Real-Time Neural Network Analysis* (Patent No. 490413-001). Controller General of Patents, Designs & Trade Marks, India. IP India.

CONFERENCE PAPERS

4. **Rakesh, V. K., Mazumdar, S., Maity, R. P., Samanta, T., Pandey, H. K., & Das, A.** (2026). Empirical Evaluation of State-of-the-Art Talking Head Generation Models. In *RAAISA 2025*, Lecture Notes in Networks and Systems, Springer. **Forthcoming**.
5. **Rakesh, V. K., Mo Moin, Sarkar, B., Mazumdar, S., Samanta, T., Pandey, H. K., & Das, A.** (2026). THGNLR: An Intelligent Application for Automated, Large-Scale Multimodal Human Audio-Video Data Acquisition and Auto-Cleansing. In *AICDAKD 2025*, Lecture Notes in Networks and Systems, Springer. **Forthcoming**.
6. **Mazumdar, S., Rakesh, V. K., & Samanta, T.** (2026). BayesFusion-SDF: Probabilistic Signed Distance Fusion with View Planning on CPU. *IEEE 5th International Conference on Intelligent Reality (ICIR 2026)*. **Accepted & presented**. arXiv:2602.19697.

AUTHORED BOOK & BOOK CHAPTERS

7. **Tyagi, A. K., & Mazumdar, S.** (2026). *Probabilistic Modelling for Advanced Data Analysis* (1st ed., Vol. 1). Elsevier. **Approved**.
8. **Mazumdar, S.** (2025). Educational Implications of 6G Technology for Society 5.0. In *Human-Centric Integration of 6G-Enabled Technologies for Modern Society* (pp. 255-265). Elsevier. doi:10.1016/B978-0-443-27434-3.00017-9.
9. **Mazumdar, S., & Chowdhury, I.** (2025). Integration of Machine Learning, Deep Learning, and Internet of Things Technologies for Stress Management and Identification: A Systematic Analysis. In *Human-Centric Integration of Next-Generation Data Science and Blockchain Technology* (pp. 407-418). Elsevier. doi:10.1016/B978-0-443-33498-6.00023-6.
10. **Hosni, H., & Mazumdar, S.** (2025). 6G-Enabled Robotics: The Future of Connectivity in Smart Industrial Manufacturing. In *Industrial Robotics in Smart Manufacturing* (pp. 241-258). CRC Press. doi:10.1201/9781003614470-12.
11. **Mazumdar, S., & Hosni, H.** (2026). Machine Learning and Deep Learning: Necessary Techniques for Extracting Meaningful Information in Large Databases. In *Next Generation Blockchain for Next Generation Society with Futuristic Technologies*. CRC Press. doi:10.1201/9781003620822-20.
12. **Rakesh, V. K., Mazumdar, S., Maity, R. P., Samanta, T., Pal, S., & Das, A.** (2026). Transparent and Trustworthy Data Practices for Talking Head Generation: A Systematic Survey of Datasets and Ethical Implications. In O. Azeroual (Ed.), *Technical Foundations and Applications of Trustworthy AI Systems* (pp. 195-224). IGI Global. doi:10.4018/979-8-2600-1433-2.ch007.

PREPRINTS

13. **Mazumdar, S., Rakesh, V. K., & Samanta, T.** (2026). PrivFedTalk: Privacy-Aware Federated Diffusion with Identity-Stable Adapters for Personalized Talking-Head Generation. arXiv:2604.08037.
14. **Mazumdar, S., & Rakesh, V. K.** (2026). TempoSyncDiff: Distilled Temporally-Consistent Diffusion for Low-Latency Audio-Driven Talking Head Generation. arXiv:2603.06057.
15. **Rakesh, V. K., Mazumdar, S., Samanta, T., Pandey, H. K., Das, A., & Pal, S.** (2026). Adaptive Video Conferencing Under Severe Bandwidth Constraints Using Audio-Driven Talking-Head Reconstruction. arXiv:2602.12758.
16. **Rakesh, V. K., Mazumdar, S., Maity, R. P., Pal, S., Das, A., & Samanta, T.** (2025). Quantitative Assessment in Talking Head Generation: A Comprehensive Review of Metrics and Loss Functions.
17. **Rakesh, V. K., Bhattacharjee, A., Mazumdar, S., Samanta, T., Pandey, H. K., Das, A., & Pal, S.** (2026). Symbolic Vedic Computation for Low-Resource Talking-Head Generation in Educational Avatars. arXiv:2602.08775.
18. **Rakesh, V. K., Mazumdar, S., Bhattacharjee, A., Samanta, T., Pandey, H. K., Das, A., & Pal, S.** (2026). Bandwidth-Aware Performance Analysis of Video Conferencing Platforms.
19. **Aftab, R., Prasad, A., Mazumdar, S., Rakesh, V. K., & Samanta, T.** (2026). RES-DARE: Failure-Aware Expert Adaptation and Rollback-Safe Self-Repair for Intrusion Detection. arXiv:2607.02687.